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Transfusion-Transmissible Infections (TTIs) among blood donors in Ethiopia's Tigray region: a study of blood bank data from 2020 to 2023

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Abstract

Introduction Transfusion-transmissible infections (TTIs) remain a major threat to blood safety, especially in resource-limited settings. In the Tigray region, where blood demand is high due to trauma, obstetric emergencies, and other medical needs, ensuring the safety of donated blood is critical. Limited data exist on the prevalence and trends of TTIs among blood donors in the region.

Method the study was conducted utilizing data from the Tigray Regional Blood Bank from 2020 to 2023. A total of 35,034 blood donations were analyzed, focusing on the prevalence of four key infections: HIV, syphilis, hepatitis B (HBV), and hepatitis C (HCV). One-way Anova, Kruskal Wallis, tests were performed to assess trends and fluctuations in infection rates over the designated period. We have used 95% level of confidence (CL) and statistical significance was declared at p-value less than 0.05.

Results The total number of donations reached 35,034, with male donors consistently outnumbering female donors across all age groups. Younger donors aged 18–34 were the most age group population who actively donate blood. Blood donation showed consistent increase from the year 2020 to 2022, but declined in 2023 (early recovery phase). Among the total blood donors, blood type A + was detected in 22.9% (8,034) followed by B + which is 22%. The overall TTI prevalence in this study was 4.33%. of those HIV was 0.257%, HBV was 3.157, HCV was 0.11 and Syphilis was 1.33%. TTIs exhibited fluctuations over the study period, rising from 4.3% in 2020 to a peak of 4.64% in 2022, before stabilizing at 4.3% in 2023. HIV cases declined steadily from 20 (0.376%) in 2020 to 13 (0.15%) in 2023. Statistical analysis revealed no significant differences were found for Total TTI, HIV, HBV, and HCV, across years or war status ($p > 0.05$). The Kruskal Wallis test indicated that no significant difference in Syphilis and overall TTI detection over time ($p > 0.05$).

Conclusion and recommendation The analysis revealed a significant decline in blood donation after the war period (early recovery phase). From the total donations in those years, male gender and young population were the

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most contributors to the blood bank. HIV, HBV, HCV, Syphilis and Total TTIs remained stable over years. These findings highlight the need for continued donor screening, and improved data collection to strengthen blood safety and ensure early detection of transfusion-transmitted infections.

Keywords Prevalence, Trend, Blood bank, Transfusion transmitted infections, Tigray, Ethiopia

Introduction

Blood transfusions are a critical component of modern medical care, yet they carry the risk of transmitting infections, known as Transfusion-Transmissible Infections (TTIs). The prevalence of TTIs among blood donors significantly impacts public health, particularly in regions like Tigray, Ethiopia. Recent data on TTIs are essential for understanding their trends and guiding public health interventions [2]. Factors such as socio-economic conditions, limited healthcare infrastructure, and a high burden of infectious diseases contribute to the prevalence of TTIs among blood donors [8]. The Tigray region has experienced increasing rates of TTIs, underscoring the need for systematic and ongoing monitoring to ensure blood safety [1].

The World Health Organization (WHO) provides guidelines for blood safety and emphasizes the importance of regular screening for TTIs to protect both donors and recipients. In Tigray, the rising number of reported cases of Hepatitis B, Hepatitis C, HIV, and syphilis among blood donors has raised concerns. Prior studies indicated a significant prevalence of these infections, which emphasizes the need for targeted interventions [2].

Many potential donors remain unaware of the risks associated with TTIs and may not be forthcoming about their health history, further complicating the situation [3]. Furthermore, while national health guidelines advocate for regular and comprehensive testing, inconsistencies in the implementation of screening protocols have been reported in the Tigray region. A 2023 analysis by Nega et al. highlighted that only a fraction of blood donations undergo thorough screening for TTIs [4]. This situation results in a critical gap in ensuring blood safety and safeguarding the health of patients who depend on transfusions.

Transfusion-transmissible infections (TTIs) remain a major threat to blood safety, especially in resource-limited settings. In the Tigray region, where blood demand is high due to trauma, obstetric emergencies, and other medical needs, ensuring the safety of donated blood is critical. Limited data exist on the prevalence and trends of TTIs among blood donors in the region. Studying TTIs will help identify the burden, guide screening strategies, improve donor selection, and strengthen blood transfusion services to reduce the risk of infection transmission and improve public health outcomes. Therefore, the study aims to provide a clearer understanding of the trends and prevalence of TTIs among blood donors in Tigray by analyzing historical blood bank data from 2020 to 2023.

Methods and material

Study setting and period

The study was conducted in the Tigray region of Ethiopia, known for its challenging health landscape exacerbated by socio-political disruptions since late 2020 [5]. The Mekelle blood bank operates under the National Blood Transfusion Service, which enforces strict screening protocols to ensure safe blood supply,this includes testing for Transfusion-Transmissible Infections (TTIs) such as HIV, syphilis, hepatitis B, and hepatitis C [6]. Blood banks are pivotal in organizing health data, providing an opportunity to assess the prevalence of TTIs through existing records [7]. The study encompasses the period from January 2020 to December 2023, which has been marked by considerable public health challenges due to both the COVID-19 pandemic and the unrest in the Tigray region [9]. This four-year time frame was chosen to provide a comprehensive analysis that captures the pre-conflict and conflict stages.

Study design and population

Evaluation of retrospective data collected through routine blood donation screening at the Mekelle blood bank, leveraging data that reflects TTIs Trend and prevalence pre and post conflict TTIs situation. In this study, data was extracted from the blood bank Monthly report for voluntary blood donors aged 18—65 years. The study population was primarily consists of individuals who donated blood at the Mekelle blood bank during the specified time period of January 2020 to December 2023. During the data collection period experts detected the TTI positivity rate using ELISA screening using make/model *Bio-Rad iMark Microplate Reader*. During the procedure, experts performed internal validation by running control samples and duplicate tests to ensure accuracy, reproducibility, and consistency of results throughout the screening process (Mekelle blood bank).

Sample size and data collection

The study comprised a total of 35,034 blood donations collected from the Mekelle blood bank over the designated four-year timeframe, with an emphasis on screening results for Transfusion-Transmissible Infections (TTIs). This estimate encompasses data from both the

pre-conflict and conflict phases, incorporating contributions from both male and female donors.

Data set and data extraction

This study utilizes a secondary dataset of voluntary blood donors consecutively collected by blood bank in the Tigray region of Ethiopia, covering January 2020 to December 2023. The dataset includes annual records of individuals tested for various TTIs (HIV, syphilis, HBV, and HCV) with results classified as positive or negative as part of blood donation screening procedure. The extraction process was conducted using a standardized protocol to ensure accuracy and consistency, with a focus on maintaining confidentiality and data integrity. Data were collated into excel spread sheet for subsequent statistical analysis, facilitating a thorough assessment of TTI trends and prevalence in the region.

Inclusion and exclusion criteria

The study included anonymized records of blood donors aged 18–65 years who donated blood at designated blood banks in the Tigray region between January 2020 and December 2023 and had complete screening results for transfusion-transmissible infections (TTIs), including HIV, Hepatitis B, Hepatitis C, and Syphilis. Records with incomplete TTI screening data or those identified as duplicates or irrelevant to the study were excluded to ensure accuracy and confidentiality.

Data management and analysis

Data management involved cleaning and organizing anonymized donor records to ensure completeness and accuracy, with exclusion of incomplete or duplicate entries. Descriptive statistics were used to summarize trends in transfusion-transmissible infection (TTI) positivity rates across years and by disease type. Assumption of normality of data was assessed using Shapiro–Wilk test for normality, Accordingly, one-way ANOVA was used for pre-war, during and post-war comparisons for normally distributed diseases, while Kruskal Wallis test was applied for non-normal distribution. Test of significance was measured at 95% level of confidence. Data analysis was employed using STATA software.

Study variables

Dependent variables

The dependent variables in this study are the specific Transfusion-Transmissible Infections (TTIs) identified among the blood donors. These include HIV, HBV, HCV, and Syphilis rates.

Independent variables

The independent variables included in the study were: Year (2020–2023) to assess temporal trends, War Status (pre-war 2020, During the war 2021 and 2022 and post-war (early recovery) 2023 for comparative analysis, Disease Type (HIV, HBV, HCV, Syphilis) as the infection category, Age (18–65 years) grouped demographic factor, Sex (male/female) to evaluate gender-based differences, and Blood Group (A, B, AB, O) as a donor characteristic.

Results

Blood donation trend in Tigray region, 2020–2023

The line graph below illustrates the trends in blood collection units from 2020 to 2023. Blood collection volumes showed a consistent upward trend from 2020 (pre-war) to 2021/22 (during war), increasing from approximately 1,000 units in 2020 to a peak of around 2,000 units in 2022, reflecting a steady growth in donations over this period. However, there was a noticeable decline in 2023 (post war), with collections dropping back to roughly 1,800 units (Fig. 1).

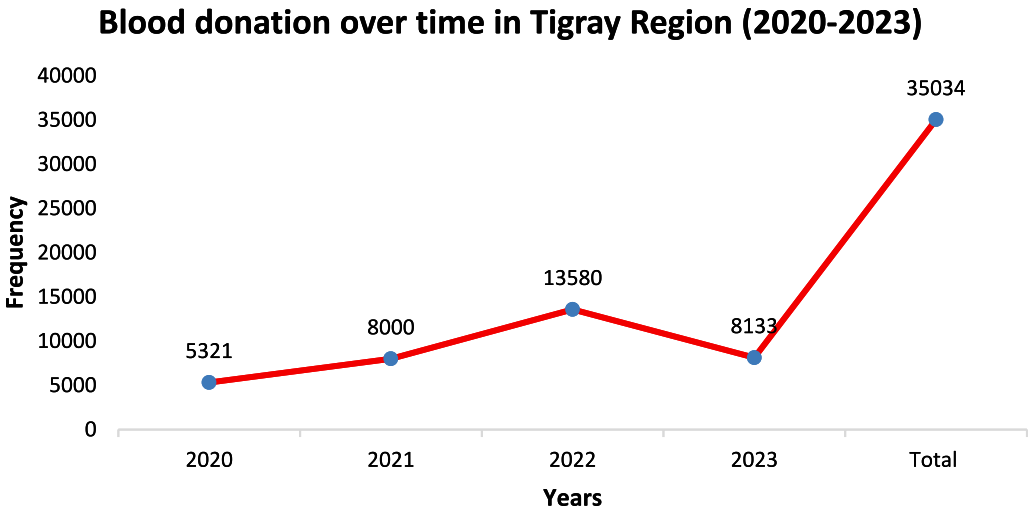


Fig. 1 Blood donation trend over time in Tigray region (2020–2023), Mekelle blood bank

Table 1 Blood donor's age and sex distribution (2020–2023)

Age group	Year	Male donors (M)	Female donors (F)	Total donors (T)
18–24	2020	1,364	1,058	2,422
	2021	1,936	850	2,786
	2022	2,617	968	3,585
	2023	989	746	1,735
25–34	2020	1,402	527	1,929
	2021	3,163	238	3,401
	2022	4,167	854	5,021
	2023	2,159	720	2,879
35–44	2020	493	105	598
	2021	1,152	188	1,340
	2022	3,194	316	3,510
	2023	1,554	874	2,428
45–54	2020	156	92	248
	2021	321	70	391
	2022	772	416	1,188
	2023	502	382	884
55–64	2020	112	12	124
	2021	71	11	82
	2022	268	8	276
	2023	190	17	207

Blood donors by age and gender in Tigray blood bank, from 2020 to 2023

This study analyzed the age and sex distribution of blood donors from 2020 to 2023, revealing a strong engagement among younger donors aged 18–34, who comprised a significant portion of total donations. The total number of donations reached 35,034, with male donors consistently outnumbering female donors across all age groups. Participation declined with increasing age, particularly in those aged 55–64, highlighting a need for targeted recruitment strategies to engage underrepresented demographics, especially females and older individuals (Table 1).

Blood donors blood group distribution in Tigray blood bank, from 2020 to 2023

The distribution of blood groups among donors from 2020 to 2023 revealed consistent patterns in the prevalence of major blood types. O + ve was the most common blood group, accounting for 37–38% of all donations across the study period, followed by A + ve (22–23%) and B + ve (21–22%), indicating a relatively stable donor population with no major shifts in overall blood group composition. The less common blood types such as AB + ve (4.8–6.1%) and Rh-negative subgroups showed minor fluctuations over time, with slightly higher proportions observed in earlier years. While year-to-year variation in percentages was minimal for most blood types, some differences were observed in AB + ve and A-ve categories (Table 2).

Trend of transfusion transited infections in among blood donors in Tigray blood bank, 2020–2023

A total of 5,321, 8,000, 13,580, and 8,133 blood units were collected and screened in 2020, 2021, 2022, and 2023, respectively. The number of donors positive for any TTI was 4.3% (229) in 2020 and 4.3% (351) in 2023. HIV cases declined steadily from 0.376 (20) in 2020 to 0.15% (13) in 2023. HBV cases were highest in 2020 at 4.21 (224), dropped to 2.6% (209) in 2023. HCV remained very low with 0.09% (5) in 2020 and only 0.01% (3 cases) in 2023. Syphilis showed an initial decline from 1.46% (78 cases) in 2020 to 0.87 (70) in 2021, followed by an increase to 1.45 (156) in 2022 and 1.54 (126) in 2023 (Fig. 2). Following checking the normality of the TTI prevalence over years, Kruskal Wallis test was conducted to detect whether there is significant difference among years. The result showed that, TTI was stable from 2020 to 2023 ($\chi^2 = 0.331$, P-value = 0.9541).

Does transfusion transmitted infections percentage increase because of war?

The analysis compared the percentage of blood donors testing positive for TTIs before (2020), during the war (2021 and 2022) and after (early recovery) 2023) the war. Shapiro–Wilk tests were conducted to assess normality, guiding the appropriate choice of statistical test. For all diseases except Syphilis, the data followed a normal distribution ($p > 0.05$), so independent t-test was applied. For Syphilis, where the data violated normality ($p = 0.034$), a Wilcoxon rank-sum test was used. The TTI infection prevalence, which is not normally distributed, shows higher values before the war, with a median of 0.92% and wider variability (IQR = 3.36). During the war, the median dropped to 0.59% with a narrower IQR (2.19),

Table 2 Comprehensive distribution of blood group among donors, 2020–2023

Year	Blood group								
	A + ve	B + ve	AB + ve	O + ve	A-ve	B-ve	AB-ve	O-ve	Total
2023	1854 (22.79%)	1742 (21.42%)	443 (5.44%)	3033 (37.28%)	231 (2.84%)	259 (3.18%)	129 (1.58%)	442 (5.43%)	8133
2022	3076 (22.61%)	3058 (22.53%)	835 (6.14%)	5062 (37.32%)	404 (2.98%)	368 (2.71%)	157 (1.16%)	620 (4.57%)	13,580
2021	1880 (23.50%)	1744 (21.80%)	384 (4.80%)	3040 (38.00%)	440 (5.50%)	128 (1.60%)	24 (0.30%)	360 (4.50%)	8000
2020	1224 (23.00%)	1171 (22.01%)	266 (5.00%)	2022 (38.00%)	266 (5.00%)	106 (2.00%)	53 (1.00%)	213 (4.00%)	5321
Total	8,034	7,715	1,928	13,157	1,341	861	363	1,635	35,034

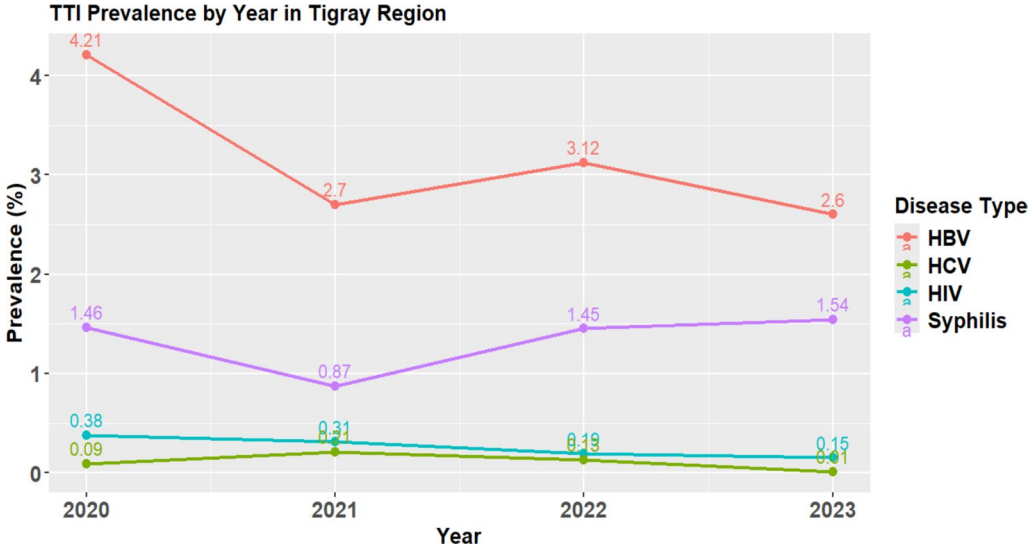


Fig. 2 Line graph showing the trend of transfusion transited infections 2020-2023, Mekelle blood bank

Table 3 Description of total TTI among blood donors by war-status in Tigray region

War-status	Mean	Std. dev	Median (IQR)	Min	Max
Pre-war	1.535	1.878235	0.92 (3.36)	0.09	4.21
During-war	1.1225	1.195046	0.59 (2.19)	0.13	3.12
Post-war	1.075	1.229051	0.845 (2.28)	0.01	2.6

Table 4 Effect of war (pre-war vs post-war) on TTI detection rates

Disease	Shapiro–Wilk p	Test used	Test statistic	p-value
Total Donors Positive for TTIs	0.5036	Kruskal Wallis Test	W = 0.281	0.86
HBV	0.628	One-way Anova	f = 8.74	0.23
HCV	0.224	One-way Anova	f = 2.75	0.39
HIV	0.9951	One-way Anova	f = 1.85	0.461
Syphilis	0.034	Kruskal Wallis Test	W = 0.35	0.76

likely reflecting reduced screening and reporting capacity. In the post-war period, the median increased slightly to 0.85% (IQR = 2.28) but remained below pre-war levels, indicating that blood safety services have not fully recovered (Table 3).

Kruskal Wallis test was run to see whether the median values of the prevalence among the war status period categories were significantly different or not. We found that there is no significant difference among categories in the variable war status ($\chi^2 = 0.281$, p-value 0.86). Among all infections, only HIV showed a statistically significant difference between pre-war and post-war periods (t = 4.34, p = 0.0492, with rate difference of 0.175% rate), indicating a meaningful decline in detection rates during the post-war period. For Total TTIs, HBV, HCV, and Syphilis, no statistically significant differences were observed ($p > 0.05$) (Table 4).

Discussion

Blood donation and Transfusion Transmission Infections (TTIs) tend in Tigray region, 2020–2023

The analysis of blood donation and transmissible infection (TTI) data in the Tigray region of Ethiopia highlights significant fluctuations in donor behavior and TTI prevalence in the periods from 2020 to 2023. Blood donation showed consistent increase from the year 2020 to 2022, but declined in 2023 (early recovery phase). During wartime, the demand for blood transfusion often surges due to mass casualties, leading to increased mobilization of donors and strengthened blood collection efforts; post-conflict, however, donation rates may decline sharply because health systems are weakened, infrastructure is damaged, and populations become displaced. This pattern has been observed in conflict settings, where blood banks report increased transfusion demand during war but struggle with supply and logistic challenges [10]. Even though there is no significant difference among years, the prevalence of TTI was immediately increased from 4.1% at the initial period of the war (2021) to 4.64% in 2022 before stabilizing back to 4.3% in 2023. Similar findings observed in other conflict-affected regions, such as South Sudan [11], indicate that wars substantially disrupt healthcare systems, leading to increases in infections like HIV and Hepatitis. Research conducted in Sierra Leone post-Ebola outbreak [12] found a rise in both syphilis and HBV among blood donors due to the breakdown of health infrastructure and limited access to screening.

In the Tigray region, although the HIV prevalence has remained relatively low, the increase in HBV cases during 2022 is consistent with trends reported in other

conflict zones where sexual violence and lack of medical resources have surged the risk of such infections among vulnerable populations [14]. The decline in HCV cases to a mere 0.01% in 2023 suggests successful public health initiatives may be underway, paralleling trends observed in Zambia 13, where targeted screening programs significantly reduced infection rates post-conflict.

Moreover, the unpredictable fluctuations in syphilis prevalence in Tigray reflect overarching themes of instability, reinforcing that ongoing public health interventions and continuous monitoring are crucial in mitigating TTI risks during recovery phases. This highlights the profound need for integrated health promotion strategies in the aftermath of conflict, focusing not just on restoring blood donation rates but also on addressing the heightened risk of infectious diseases.

TTIs case detection rate among blood donors in Tigray blood bank, 2020–2023

The findings from the Tigray region present a nuanced landscape of transfusion-transmissible infections (TTIs) that emphasizes stability amidst the backdrop of conflict. Prevalence rates remained consistent at approximately 4.3% both before and after the conflict in 2023, demonstrating resilience despite significant disruptions. This stability contrasts with neighboring countries, such as Uganda, where TTI prevalence was observed at 6.4% [15], and Kenya, which reported a much higher rate of 14.1%, primarily driven by Hepatitis B (5.6%) [16]. Research conducted in Sudan in 2019 indicated that the overall prevalence of TTIs among blood donors in Port Sudan Central Blood Bank over 3 months of 2017 was 103/513 (20.1%) [17]. In addition, the Sero-prevalence for TTI markers were 0.18% HIV, 2.58% HBV, 0.57% HCV and 0.49% Syphilis in Eritrea [18].

Comparative data further emphasize the regional variability in transfusion-transmissible infections (TTIs) within Ethiopia. In the Tigray region, the sero-prevalence rates for Hepatitis B (HBV) are reported at 2.4% (CI: [1.8%, 3.0%]), with HIV and HCV rates at 0.4% (CI: [0.2%, 0.6%]). In stark contrast, areas such as Hossana have reported TTIs as high as 14.38% (CI: [12.5%, 16.3%]) (Beykaso et al., 2021), while Bahir Dar shows a prevalence of 6% (CI: [4.8%, 6.1%]) [19]. Furthermore, Bule Hora presents a TTI prevalence of 4.6% (CI: [3.8%, 5.5%]) [20]. These disparities underscore the necessity for targeted public health strategies, as the infection burden varies significantly across different geographic regions.

Trends over time indicate a general decline in TTI prevalence throughout Ethiopia, with several studies noting reductions in rates. A retrospective analysis from 2001 to 2016 reported a decline in the Tigray region, aligning with improvements in blood safety measures, and other Ethiopian studies demonstrated a decrease from 5.7% in 2014 to 3.4% in 2019 [21]. This downward trend mirrors similar findings in Sudan and Rwanda, where advancements in screening techniques and robust public health initiatives contributed to reduced prevalence rates [22, 23]. Notably, the shift from replacement to voluntary blood donation schemes has also been associated with lower infection rates, reinforcing the value of improved donor selection processes [24] (Patil et al., 2020).

While the Tigray region shows a declining trend in HBV, from 4.21% in 2020 to 2.6% in 2023, other studies have revealed more variable outcomes, with some regions experiencing fluctuations in positivity rates. For example, Dessie reported sero-prevalence rates around 6.25%, indicating that not all areas are benefiting equally from blood safety improvements [25, 26].

The discrepancies in findings regarding the prevalence of transfusion-transmissible infections (TTIs) in Ethiopia can be attributed to several factors, including geographic variability, differences in blood donation practices (e.g., voluntary vs. replacement), and variations in public health infrastructure and resources [27, 28]. Local epidemiology, levels of awareness about blood safety, and the impact of conflict on healthcare access also influence TTI rates [29]. Furthermore, variations in study design, methodology, socioeconomic conditions, and regulatory environments contribute to inconsistencies in prevalence rates [30, 31]. Addressing these challenges requires targeted public health strategies, enhanced surveillance, and ongoing investment in healthcare systems and education.

Overall, despite the stabilization of TTI prevalence in Tigray, the broader trends across Ethiopia highlight the ongoing challenge of geographic disparities in infection rates. Continuous monitoring and targeted public health interventions are essential to address these inconsistencies and further reduce the burden of TTIs nationally. Investment in blood safety initiatives and donor education remains crucial to maintain and enhance the observed stability in the Tigray region post-conflict.

In comparing the overall incidence rate of transfusion-transmissible infections (TTIs), the rates exhibited slight fluctuations, with an approximate rate of 4.2% in 2020 ([3.77%, 4.85%] CI), peaking at 4.5% in 2022 ([4.29%, 4.99%] CI), and returning to 4.2% in 2023 ([3.88%, 4.76%] CI). These findings are consistent with trends observed in other East African countries, where TTI rates have similarly shown stability despite fluctuations [32]. Notably, hepatitis B virus (HBV) displayed the highest incidence among individual infections, beginning at 4.2% in 2020 ([4.12%, 4.76%] CI) and decreasing to 2.6% by 2023 ([2.43%, 2.77%] CI), aligning with findings from a study in Uganda that reported a substantial decline in HBV prevalence over the same period [33]. Syphilis rates remained stable, ranging between 1% and 1.5% throughout the

years ([0.87%, 1.54%] CI in 2023), similar to previous studies in the region that highlighted consistent rates despite occasional spikes [34]. In contrast, both HIV and hepatitis C virus (HCV) maintained the lowest incidence rates, consistently below 0.5% ([0.15%, 0.41%] CI for HIV in 2023; [0.01%, 0.21%] CI for HCV in 2023). This reflects trends seen in multiple East African studies reporting low but stable rates for these infections [35]. Overall, the observed trend towards stabilization or slight decline in TTI rates aligns with regional data, indicating improved health interventions and awareness, as supported by recent research in East Africa [36].

Conclusion and recommendations

The overall TTI prevalence in Mekelle blood bank remained significantly unchanged over time. The young age group and male gender was the most contributors during these periods. Out of the total donations A + blood type was detected 22.9% of them. The overall TTI prevalence in this study was 4.33%. of those HIV was 0.257%, HBV was 3.157, HCV was 0.11 and Syphilis was 1.33%. It is crucial to implement ongoing monitoring, focused public health programs, and enhanced investment in healthcare systems and education. Additionally, there should be a strong emphasis on integrated health promotion strategies to strengthen recovery efforts.

Limitation and strength of the study

Limitations: The study's retrospective design may introduce biases from incomplete data, and its focus on blood donors may not represent the wider population, potentially leading to under reported TTI prevalence. Generalizability is limited to the Tigray region, and external factors such as stigma, political conflict, and the COVID-19 pandemic could confound results.

Strengths: The study boasts a large sample size that enhances statistical reliability and utilizes real-world data to provide insights into TTI prevalence. It targets vulnerable populations, employs comprehensive screening for multiple TTIs, and offers historical context for understanding trends. Additionally, the findings can inform health policies and contribute to the existing literature on TTIs in underserved areas.

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Author contributions

Demoz K.: Conceived and designed the study, drafted and revised the manuscript. Gessesse T., Asfah M.: Provided infectious disease expertise, contributed to literature review and interpretation and revisions. Hagos S., Baraki K., Kahsay W., Meresa D., Hadgu T.: Conducted statistical analysis, data interpretation, and manuscript formatting. Kahsay H.: Project management, team coordination, administrative support, and final approval. All authors reviewed and approved the final manuscript and are accountable for its content.

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Data availability

The data that underpins the findings of this study has been stored in the archive of the Tigray Regional Blood Bank and is also included in the article. For additional inquiries, please reach out to the corresponding author at kirosdz5@gmail.com.

Declarations

Ethics approval and consent to participate

The Study was conducted with a strong focus on ethical standards. The research exclusively utilized secondary data from anonymized blood bank records, thus ensuring no direct interaction with human subjects. An Institutional Review Board (IRB) review was sought, leading to the conclusion that formal ethical approval was not necessary due to the minimal risk associated with the research and the de-identified nature of the data. The study complied with ethical guidelines set forth by national and international bodies, including the Declaration of Helsinki and the Belmont Report, emphasizing confidentiality and beneficence. This commitment to ethical principles ensures the integrity of the research while promoting public health insights in the context of TTIs in the Tigray region.

Consent for publication

The authors provided their consent for publication, and all of them ethically approved this original manuscript.

Competing interests

The authors declare no competing interests.

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